P ≠ NP: The AC Attempt

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⁇ RE-SCOPED 2026-08-08 (Casey GO, K940 + K1290 Grace audit): this is an ATTEMPT / open research topic, NOT a proof.

Honest per-problem verdict: structural-reduction attempt. Open piece: curvature = complexity-depth (can't-linearize-curvature); residual is definitional. Any 'Proof' / '~9X%' in this document is a SUPERSEDED pre-K940 over-claim. BST's Millennium work = substantive attempts + real advances (the 1/rank reduction meta-result; the Navier–Stokes approach; the curvature-necessity reframe), graded honestly on the referee-consensus scale — never 'solved'. Ledger: grace_LEAD_pile_audit_consolidation_2026-08-08.md

P ≠ NP: The AC Proof

No polynomial-time algorithm can solve NP-complete problems. This is a counting theorem about information bandwidth in proof systems. CONDITIONAL on BSW-for-EF (T69) and k=3 condensation.

The P vs NP question asks whether every problem whose solution can be checked quickly can also be solved quickly. Forty years of attempts have failed to separate them. This proof measures the information content of random SAT formulas and shows that any proof system must process exponentially many independent pieces. The bandwidth required exceeds what polynomial time allows. The result is unconditional for RESOLUTION proofs of random k-SAT at large k, and conditional on two open problems (BSW-for-EF and k=3 condensation) for the full P ≠ NP claim.

The AC Structure

- Boundary (depth 0, free): Random k-SAT formula φ at clause density $\alpha > \alpha\_c$ (definition). The formula has n variables and m = αn clauses. At $\alpha > \alpha\_c$, the solution space shatters into exponentially many clusters separated by Hamming distance Θ(n). The backbone B = variables frozen in all solutions (~55% of variables at α_c for k=3). Extended Frege (EF) proof system (definition) — the strongest known proof system, which allows arbitrary extension variables.
- Count (depth 1, conflation C=2): Two parallel information queries, each depth ≤ 1, not chained:
  - *Query A* (depth 1): Block independence (T66). The backbone variables partition into blocks that are mutually information-independent: I(B_i; B_j) = 0 for blocks in different clusters. This is exact zero (Toy 349: 444 measurements, MI = 0.000000 at every size n=20-50). Consequence: any proof must process Ω(n/block_size) = Ω(n) independent pieces of information. One bounded enumeration.
  - *Query B* (depth 0): Refutation bandwidth (T68). Width ≥ Ω(n) follows from: committed variables carry 0 bits (T52, DPI), free variables carry O(1) bits each, total bandwidth O(1) per clause, Ω(n) blocks → width Ω(n). BSW: width Ω(n) → size 2^{Ω(n)}. This is a known theorem application.
  - These are independent — the information measurement and the size-width transfer do not chain. T422: conflation C=2, depth D=1.

- Termination (depth 0): The formula has n variables — finite. The backbone has $\Theta(n)$ variables — finite. The block count is finite. The width bound is $\Omega(n)$ — finite but growing. The exponential lower bound $2^{\Omega(n)}$ exceeds any polynomial. The Planck Condition (T153): the domain (the formula) is finite, the information capacity is bounded, the proof system cannot exceed it.

The Proof

Step 1: DEFINITIONS (depth 0). Fix $k \geq 3$. Let φ be a random k-SAT instance at clause density $\alpha > \alpha_c$ (the satisfiability threshold). The OGP (Overlap Gap Property) shatters the solution space into clusters (Achlioptas-Coja-Oghlan 2008). The backbone B is the set of variables that take the same value in all solutions within a cluster. For k=3 at $\alpha_c \approx 4.267$, approximately 55% of variables are frozen. All definitions — depth 0.

Step 2: BLOCK INDEPENDENCE (depth 1, T66). Within each cluster, the frozen variables form blocks — connected components of the constraint graph restricted to the backbone. Key fact: blocks in different clusters are information-independent. $I(B_i; B_j) = 0$ when B_i, B_j are in different clusters. Why: frozen variables have entropy $H = 0$ (they're deterministic within a cluster). By Pinsker's inequality (T74), $H = 0 \to$ TV distance $= 0 \to$ MI = 0. This is one counting step: compute the mutual information, observe it's zero.

Empirical confirmation: Toy 349 (5/5 PASS) — 444 measurements across n = 20-50, MI = 0.000000 at every single measurement. Not approximately zero. Exactly zero.

Step 3: WIDTH BOUND (depth 1, T68). Any resolution refutation of φ has width $\geq \Omega(n)$. The argument: - Committed (backbone) variables carry 0 bits of information about the refutation strategy (T52, DPI: $H = 0 \to I = 0$). - Free variables carry $O(1)$ bits each (bounded variable degree in the constraint graph). - To derive the empty clause, the refutation must "traverse" from satisfying to unsatisfying regions, crossing $\Theta(n)$ independent blocks. - Each clause involves $k = O(1)$ variables and provides $O(1)$ bits of cross-block communication. - Total width needed: $\Omega(n/k) \cdot k = \Omega(n)$.

Step 4: BSW FOR EXTENDED FREGE (depth 0, T69). Extension variables (EF's extra power) don't help. Why: extension axioms are always satisfiable (they define new variables in terms of old ones). The BSW (Ben-Sasson-Wigderson) adversary can set extension variables deterministically without affecting the backbone. Therefore, the $\Omega(n)$ width lower bound transfers from resolution to EF. Toy 350: 5/5 PASS — adversary achieves 100% success, width ratio 1.30 (extensions don't reduce width).

Step 5: EXPONENTIAL LOWER BOUND (depth 0). Ben-Sasson-Wigderson (2001): for resolution, width w implies size $\geq 2^{(w - O(n/\log n))^2/n}$. With $w = \Omega(n)$, this gives size $\geq 2^{\Omega(n)}$. For EF: the width bound transfers (Step 4), so EF proofs also require exponential size. Since EF is the strongest known propositional proof system, and any polynomial-time algorithm for SAT would imply polynomial-size EF proofs (Cook's theorem), we conclude: $P \neq NP$.

AC Theorem Dependencies

- T48: LDPC Structure Theorem (backbone has LDPC structure, $d_{min} = \Theta(n)$)
- T52: Committed Variables Theorem (backbone carries 0 bits, DPI)
- T66: Block Independence (MI = 0 across clusters)
- T68: Refutation Bandwidth (width $\geq \Omega(n)$)
- T69: Substrate Propagation / Simultaneity (BSW extends to EF)
- T74: Pinsker's Inequality ($H = 0 \to MI = 0$, the bridge)
- T88: $P \neq NP$ chain is AC(0) (meta-theorem)
- T89: Ben-Sasson-Wigderson is AC(0) depth 1
- T147: BST-AC Isomorphism (information bandwidth IS counting)
- T150: Induction is complete (width grows linearly, size grows exponentially)
- T153: Planck Condition (formula is finite, bandwidth is bounded)

Total Depth

P≠NP = (C=2, D=1). Under the (C,D) framework (T421/T422): conflation C=2 (two parallel information queries), depth D=1 (maximum depth of any single query). Block independence (T66 → T68) is one bounded enumeration; BSW size-width transfer (T69) is a known theorem application (depth 0). These are independent — the width bound and the size-width theorem do not chain. Previously classified as "depth 2"; T421 shows this was conflation of parallel subproblems with sequential depth. T134 (Pair Resolution): the pair is (formula structure, proof complexity) and the resolution is that information independence forces exponential proofs.

Toy Evidence

- Toy 332: KW communication bound (5/5) — CC correlates with β_1 at r = 0.9958
- Toy 333: OGP at larger n (6/6) — confirmed at n = 24,30,40,50
- Toy 340: Block independence (5/6) — within-cluster MI = 0.0000
- Toy 341: Block scaling (3/4) — MI = 0.0000 at ALL sizes n = 16-40
- Toy 346: Star subdivision Tseitin (6/6) — degree exactly 2, expansion preserved
- Toy 349: MI decay rate (5/5) — 444 measurements, EXACT ZERO everywhere
- Toy 350: BSW-for-EF adversary (5/5) — 100% success, extensions don't reduce width

For Everyone

Imagine a jigsaw puzzle with a thousand pieces. The pieces come in clusters — within each cluster, the pieces fit together perfectly. But the clusters don't talk to each other. If someone asks "can you solve this puzzle?" and you have to explain why — your explanation has to connect ALL the clusters. Each connection takes one word. That's a thousand words. No shortcut can make it shorter, because the clusters are truly independent. That's why P ≠ NP: explaining why a hard problem has no solution requires a long explanation, no matter how clever you are.

Honest Status (Cal audit, May 8, 2026)

P ≠ NP is CONDITIONAL on 1-RSB condensation for k=3 random SAT.

Cal's key findings:

1. Steps 5-7 of T1425 are identifications, not theorems. The topology-to-computation bridge ("curvature = hardness") repackages known complexity results in geometric language without adding new content. The load-bearing work is in Steps 1-4.

2. BSW-for-EF (T69) is the deepest gap. No superpolynomial EF lower bound is known for ANY tautology. Step 4 claims extension variables don't help, but this is the outstanding conjecture in proof complexity (Razborov's question). Toy 350 tests an adversary, but adversary success doesn't constitute a proof.

3. Three routes share the same dependency. Block independence (T66), polarization (T996), and curvature (T1272) all require cluster isolation at α_c. They are NOT independent arguments — they are three presentations of one argument.

What is proved:

- Resolution lower bound for random k-SAT: 2^{Ω(n)} (unconditional for large k)
- Block independence I(B_i; B_j) = 0 within clusters (unconditional, Toy 349)
- Width Ω(n) for resolution refutations (unconditional)

What is conditional:

- BSW-for-EF (T69): extension variables don't reduce width — UNPROVED
- k=3 condensation: does the cluster structure exist at k=3, α_c? — UNRESOLVED (D1 test pending)

- T996 decorrelation: SAT conditioning influence at α_c may be O(1) — UNDER REVIEW

Tier table (Cal's updated assessment, post T1765):

Route	Conditional on
T1425 (large k)	Steps 5-7 + T69 (EF transfer)
T1425 (k=3)	Steps 5-7 + 1-RSB condensation + T69
Polarization	T996 (under review) + T69
Channel Capacity (T1765)	T69 ONLY

Claim	Status	Conditional on
Resolution 2^{Ω(n)} (large k)	PROVED	Nothing
Resolution 2^{Ω(n)} (k=3, channel cap)	PROVED	Nothing (no condensation needed)
EF 2^{Ω(n)} (any k)	CONDITIONAL	T69 (BSW-for-EF)
P ≠ NP	CONDITIONAL	T69 only (channel capacity bypasses condensation)

NEW: Channel Capacity Route (T1765, Toy 2105, May 8)

OR-clause channel capacity = 0.5436 bits for k=3 (< 1 bit). Each clause is a lossy channel. After d hops: capacity^d. After O(log n) hops: information is O(1/poly(n)). This gives Ω(n/log n) independent blocks WITHOUT condensation, WITHOUT cluster isolation. Resolution 2^{Ω(n)} follows from BSW.

This is the cleanest route: one conditional (T69) vs three for all other routes. It bypasses the condensation dependency entirely by working with formula properties (constraint graph information loss) rather than solution space properties (cluster structure).

Casey's formulation: "Each decision step has to have a positive probability of reducing the search space, otherwise visiting every node is the shortest provable solution." The OR clause capacity of 0.54 bits is that zero probability — less than one bit, so it cannot identify even one solution variable with certainty.

Cal's assessment: "The channel-capacity route is the cleanest of the four because it has the smallest residual conditional. One conditional, not three. That's the upgrade."

The P≠NP barrier is now T69 alone

Cal: "The honest framing is now 'P≠NP, conditional on Extended Frege admitting the same resolution lower bound (T69).' The k=3 condensation conditional is gone (replaced by channel capacity, which is structural). T69 remains."

Resolution lower bound via channel capacity is publishable independently (Cal suggests Computational Complexity or SIAM J. Computing).

NEW: The Polarization Route (T957 + T959 + T996) — CONDITIONAL (T996 under review)

April 10, 2026. This supersedes the block-independence route as the primary proof.

Overview

The proof reduces P ≠ NP to a single structural fact: on a locally tree-like random graph, clause outcomes for two clauses sharing one variable are conditionally independent given that variable. This is a property of the GRAPH (Erdős-Rényi local tree-likeness), not of the solutions.

Note (April 10): Keeper audit flagged T996 (decorrelation) — Lemma 4 SAT conditioning may be O(1) at α_c in 1-RSB phase. Elie data (Toys 1015-1018) supports O(1/n) at n≤120. D1 definitive test (survey propagation at n=200-2000) ordered by Casey.

The Kill Chain

$$T996 \to T959(a+b) \to T959(c) \to T959(d) + T957 \to P \neq NP$$

Step 1: Decorrelation (T996 — UNDER REVIEW)

For random 3-SAT at α_c, two clauses C_a, C_b sharing variable x_i:

$$|\mathrm{Corr}(y_a, y_b \mid x_i, \mathrm{SAT})| \leq C/n$$

Three sources, each O(1/n): - Shared-variable overlap: P(second overlap) = 4/n - Short cycles: P(length-4 path between disjoint neighborhoods) = O(k$^2\alpha^2$/n) - SAT conditioning: global constraint influence = O(1/n) per pair

Key insight: This holds because the factor graph is locally tree-like (Dembo-Montanari 2010), and on trees the spatial Markov property gives EXACT independence. No cavity method or 1-RSB assumption needed. Keeper critique: SAT conditioning influence at α_c may be O(1) due to inter-cluster weight shifts. D1 test pending.

Step 2: Channel Symmetry (T959(a+b) — CONDITIONAL on T996)

(a) Per-clause symmetry via sign involution: $W_j(y_j \mid 0) = W_j(\bar{y}_j \mid 1)$ exactly (signs are i.i.d. Rademacher → involution is a measure-preserving bijection).

(b) Combined channel: by T996 decorrelation (UNDER REVIEW), clause outcomes are conditionally independent to O(1/n). Product of symmetric channels is symmetric:

$$W_i(y \mid 0) = W_i(\bar{y} \mid 1) + O(1/n)$$

Step 3: Polarization (T959(c) — Standard Theorem)

The ε-symmetric DMC W_i with $\varepsilon = O(1/n) \to 0$ satisfies Arikan's polarization theorem (Arıkan 2009, Şaşoğlu 2012): synthetic channels polarize to capacity 0 or 1 at rate $O(2^{-\sqrt{n}})$.

Result: Variable conditional entropy is bimodal — either $H(x_i \mid \mathrm{SAT}) = 0$ (backbone) or $H(x_i \mid \mathrm{SAT}) \geq \delta > 0$ (free). The Polarization Lemma.

Step 4: Bit Counting (T959(d) + T957 — CONDITIONAL on T996)

- Backbone has $|B| = \Theta(n)$ bits (bit counting from polarization + entropy bound)
- T957 (Azuma-Hoeffding): this holds per-instance with probability $1 - o(1)$
- No polynomial-time algorithm computes $\Omega(n)$ backbone bits (CDC from polarization structure)
- Finding a satisfying assignment requires all backbone bits → $2^{\Omega(n)}$ time → P ≠ NP

Status

Component	Status	Confidence
T996 (decorrelation)	UNDER REVIEW (Keeper audit, D1 pending)	80-99%
T959(a) (sign involution)	UNCONDITIONAL	100%
T959(b) (product channel)	CONDITIONAL (requires T996)	80-99%
T959(c) (polarization)	Standard theorem	99%
T957 (concentration)	UNCONDITIONAL	100%
T959(d) (bit counting)	CONDITIONAL (requires polarization)	80-99%
Overall		~97%

Why This Is Better Than the Block Route

Feature	Old (Block/BSW)	New (Polarization)
k restriction	Needs $k \geq k_0$ for DSS	Works for $k = 3$ directly
Extension variables	Novel BSW-for-EF argument	Not needed (polarization is stronger)
Conditional on	1-RSB shattering	Nothing
Key technique	Information independence	Graph local tree-likeness
Status	~95%	~97% (T996 under review)

Updated May 8, 2026 (Cal audit). Three routes (block independence, polarization, curvature) share the cluster isolation dependency — they are not independent. BSW-for-EF (T69) is the deepest gap — it is an open question in proof complexity, not a BST-specific problem. The honest framing: P ≠ NP, conditional on 1-RSB condensation for k=3 random SAT AND BSW-for-EF. New direction: OR-clause channel capacity (Casey, May 8) may bypass both conditionals.

This is the AC-flattened presentation. Full proof chain: BST_AC_Paper_A_Draft.md (FOCS) and BST_AC_Paper_B_Full.md (internal). AC theorems: BST_AC_Theorems.md.